

Solution Requirements

Date	05 July 2026
Team ID	
Project Name	Health History Organizer
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

Create a comprehensive list of requirements to define exactly what the solution will do and how well it will perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users shall securely log in using email and password with encrypted credentials and JWT-based authentication.	High
2	Authorization Levels	Patients manage their own records while administrators manage users and monitor the system using role-based access.	High
3	External Interfaces	Integrate with FastAPI REST APIs, SQLite database, AI health summary services and email notification services.	High
4	Transaction Processing	Support Create, Read, Update and Delete (CRUD) operations for health records while maintaining data integrity.	High
5	Reporting	Generate health summaries, downloadable PDF reports, dashboards and searchable medical history.	Medium

6	Business Rules	Validate mandatory fields, prevent duplicate records and maintain modification history for each patient.	High
7	Compliance to Laws or Regulations	Protect sensitive health data using secure authentication, encryption and user consent mechanisms.	High
8	Other	Provide AI-assisted health summaries, advanced search, secure backup and responsive access.	Medium

Step 2: Non-Functional Requirements (NFR)

Specify the quality attributes and measurable criteria that define how the Health History Organizer should perform.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should load pages quickly and process API requests efficiently.	Average page load time under 2 seconds.
2	Scalability	The system should support increasing numbers of users and medical records without major performance degradation.	Supports at least 10,000 registered users.
3	Security & Data Privacy	Protect medical records using encryption, secure authentication, and controlled access.	AES-256 encryption, JWT authentication, HTTPS communication.
4	Reliability & Availability	Ensure stable service with minimal downtime and reliable database backups.	99.9% monthly uptime with automated daily backups.
5	Usability & Accessibility	Provide an intuitive, responsive interface accessible across devices and	Responsive UI with WCAG-inspired accessibility practices.

		browsers.	
6	Maintainability & Portability	Code should be modular, documented, and easy to deploy across development and production environments.	Modular FastAPI architecture with documented APIs and version control.